

Report on Adaptive Weighted Sequential GDS-ARM

Factor Screening in Mixed-Level Supersaturated Designs

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Abstract. This report studies adaptive extensions of mixed-level GDS-ARM for factor screening in supersaturated designs. The baseline repeatedly fits Gauss–Dantzig selector models on random interaction subsets and aggregates the best models. Three extensions are evaluated: BIC-weighted model aggregation, sequential stopping when inclusion scores stabilize, and evidence-guided interaction-block sampling. The benchmark uses three mixed-level design templates, 18 design–scenario conditions, and 200 simulated datasets per condition. Under a common configuration, the original baseline obtains the highest mean score, 0.601. Weighted aggregation raises power but also increases false selections. Adaptive stopping reduces the mean repetition budget from 1000 to 368.81 and runtime from 216.21 to 79.93 seconds while remaining close to the weighted method on score. Adaptive interaction sampling produces localized gains but no overall improvement. The evidence supports adaptive stopping as a computational enhancement while retaining the original baseline when false-positive control and total score are primary.

Reference method	Mixed-level GDS-ARM of Zhang, Singh, and Stufken (2026)
Study scope	Baseline tuning, weighted aggregation, adaptive stopping, adaptive interaction sampling, and unified comparison
Evidence base	Three design templates, 18 design–scenario conditions, and 200 Monte Carlo datasets per condition
Principal finding	Adaptive stopping gives the clearest efficiency gain; the original baseline remains the best overall scorer

1 Introduction

Supersaturated designs are used when the number of candidate effects exceeds the number of experimental runs. Full-model estimation is then impossible without a sparsity assumption, so the task becomes one of screening: identify a small set of important factors while controlling false selections. Mixed-level designs are especially difficult because a factor with q_i levels contributes $q_i - 1$ contrast columns and a two-factor interaction contributes $(q_i - 1)(q_j - 1)$ columns. The candidate interaction space can therefore grow much faster than the number of physical factors.

The Dantzig selector provides a sparse solution in high-dimensional regression [1]; the Gauss–Dantzig selector adds active-set extraction and ordinary least-squares refinement [2]. GDS-ARM stabilizes this approach by aggregating selections over random submodels [3]. The mixed-level extension of Zhang et al. [4] supplies the baseline used here.

The present study asks whether three rigid stages of that baseline can be made adaptive. First, equal top-model counts are replaced by BIC evidence weights. Second, the fixed repetition budget is replaced by a stability-based stopping time. Third, uniform interaction sampling is replaced by a block-level proposal that combines direct interaction evidence with a heredity signal from parent main effects. The goal is not merely higher power: a useful modification must balance sensitivity, false selections, and computational cost.

2 Methodology

2.1 Baseline mixed-level GDS-ARM

For a centered response and contrast-coded model matrix,

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon},$$

each baseline repetition combines all main-effect columns with n_{int} uniformly sampled interaction columns. A Dantzig path is solved, active sets are extracted by clustering coefficient magnitudes, and post-OLS BIC ranks the candidate models. After n_{rep} repetitions, the best n_{top} models are aggregated by inclusion frequency. Columns appearing in at least a p_{keep} fraction of these models enter a final bidirectional stepwise regression. Selected columns are mapped back to effects and then to physical factors.

2.2 BIC-weighted aggregation

Let $B_{(1)}, \dots, B_{(T)}$ be the top models ordered by BIC. The weighted method assigns

$$w_t = \frac{\exp\{-\frac{1}{2}(\text{BIC}_{(t)} - \text{BIC}_{(1)})\}}{\sum_{s=1}^T \exp\{-\frac{1}{2}(\text{BIC}_{(s)} - \text{BIC}_{(1)})\}}, \quad \pi_c = \sum_{t=1}^T w_t \mathbf{1}\{c \in B_{(t)}\}. \quad (1)$$

Column c is retained when $\pi_c \geq \tau$, with $\tau = p_{keep}$ in the reported comparisons. Equal BIC values recover ordinary top-model frequencies, so the weighted rule is a direct generalization of baseline aggregation.

2.3 Adaptive repetition stopping

Repetitions are processed in batches. After batch b , score stability is measured by

$$\Delta_b = \left\| \boldsymbol{\pi}^{(b)} - \boldsymbol{\pi}^{(b-1)} \right\|_{\infty}. \quad (2)$$

The procedure stops after $R_{\min}$ repetitions when $\Delta_b < \varepsilon$ for K consecutive batches, subject to a maximum $R_{\max}$. The benchmark uses batch size 50, $R_{\min} = 100$, $R_{\max} = 1000$, $\varepsilon = 0.01$, and $K = 2$. This rule targets a stable evidence vector rather than a predetermined computational budget.

2.4 Adaptive interaction sampling

For interaction block $e = (a, b)$, let v_e denote accumulated interaction evidence and u_a, u_b the parent main-effect scores. The next-batch block weight is proportional to

$$\tilde{q}_e = q_{\min} + \frac{1 - \lambda}{|\mathcal{E}_{\text{int}}|} + \lambda \{(1 - \eta)v_e + \eta u_a u_b\}. \quad (3)$$

The normalized block probability is distributed uniformly across its columns. Two uniform warm-up batches prevent early over-concentration. The tested values are $\lambda = 0.7$, $\eta = 0.5$, and $q_{\min} = 0.001$.

3 Simulation Study

3.1 Designs and scenarios

Table 1 summarizes the three generated design templates. They match the dimensions and factor-level patterns of the reference designs, but they do not reuse the exact supplement rows.

Table 1: Mixed-level benchmark design templates.

Design	Runs n	Factors m	Level pattern	Main columns	Interaction columns	Ratio
1	21	8	$2^4 3^4$	12	62	0.571
2	27	20	$2^{10} 3^{10}$	30	425	1.111
5	8	12	$2^8 4^4$	20	178	2.500

Each design is evaluated under six sparse scenarios, for 18 conditions in total. Active effects follow a weak-heredity construction. Within an active multi-column effect, a nonempty subset of columns is selected; nonzero coefficients have random signs and magnitudes from $N(5, 1)$, and $\boldsymbol{\varepsilon} \sim N(\mathbf{0}, I_n)$. Each condition contains 200 Monte Carlo datasets, and matched seeds are reused across methods when possible.

3.2 Common settings and evaluation

The unified comparison fixes $n_{\text{int}} = 2n$, $n_{\text{top}} = 20$, $p_{\text{keep}} = 0.25$, and $n_{\text{rep}} = 1000$ for fixed-budget methods. All methods use the same Dantzig path, post-OLS BIC ranking, and final bidirectional stepwise thresholds. At the factor level,

$$\text{Power} = \frac{|\hat{T} \cap T|}{|T|}, \quad \text{Error} = \frac{|\hat{T} \setminus T|}{m - |T|}, \quad \text{Score} = \text{Power} - \text{Error}. \quad (4)$$

4 Results

4.1 Baseline tuning

The interaction budget is strongly design-dependent. On Design 1, mean step-level score rises from 0.621 at $0.5n$ to 0.813 at $3n$. On Design 2 it falls from 0.556 at $0.5n$ to 0.382 at $6n$. The $(n_{\text{top}}, p_{\text{keep}})$ sweep places the best mean score, 0.607, at $(60, 0.25)$, essentially tied by $(20, 0.40)$. The common setting $(20, 0.25)$ scores 0.595 but retains higher power, 0.832.

The repetition sweep gives scores of 0.592, 0.596, 0.600, and 0.597 at $n_{\text{rep}} = 100, 500, 1000, 4000$, while runtimes increase from 21.57 to 107.79, 215.68, and 863.06 seconds. Thus $n_{\text{rep}} = 1000$ is the strongest tested fixed score–cost compromise. Table 2 and Figures 1–2 summarize all baseline tuning plots.

Table 2: Baseline tuning conclusions.

Study	Main result	Implication
<i>nint</i>	Opposite score trends on Designs 1 and 2	Use $2n$ as a common comparison point, not a universal optimum.
$(ntop, pkeep)$	Score frontier near $pkeep = 0.25\text{--}0.40$	Higher $pkeep$ controls error but sacrifices power.
<i>nrep</i>	Score peaks at 1000; 4000 costs four times more	Additional repetitions have sharply diminishing returns.

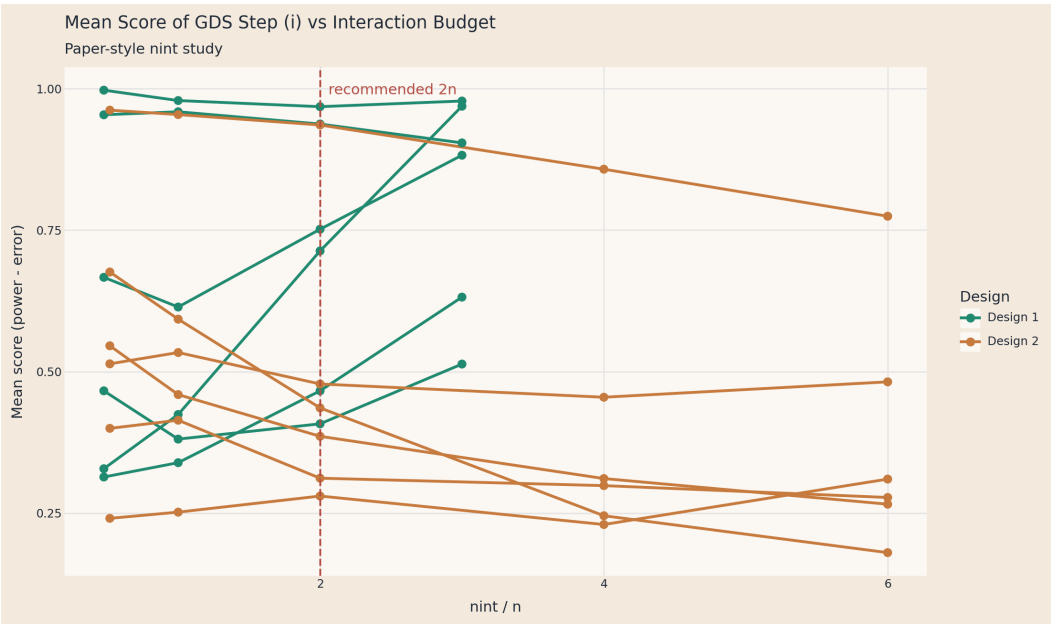
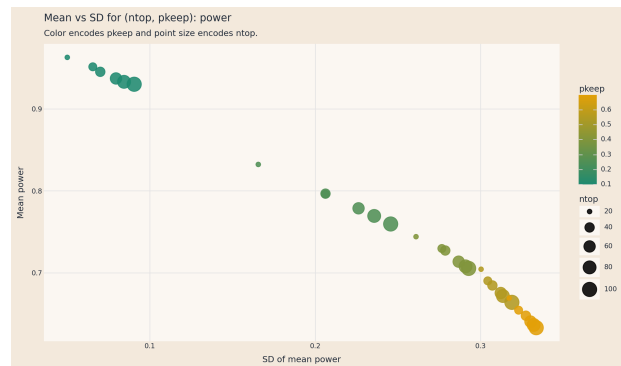
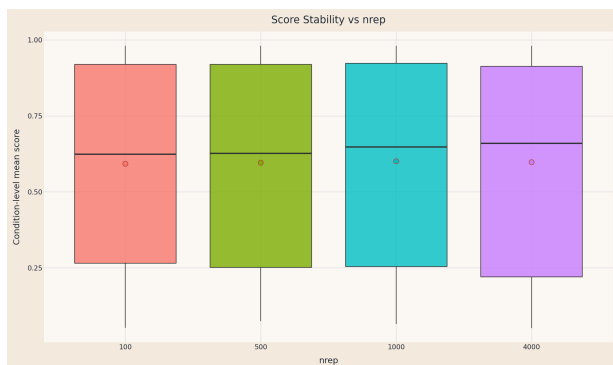
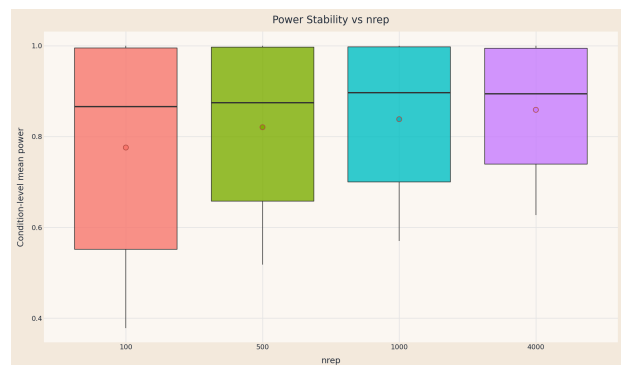


Figure 1: Baseline interaction-budget study. The effect of increasing *nint* differs sharply across the two design templates.

(a) Score frontier over $(ntop, pkeep)$ (b) Power frontier over $(ntop, pkeep)$ 

(c) Score across fixed repetition budgets



(d) Power across fixed repetition budgets

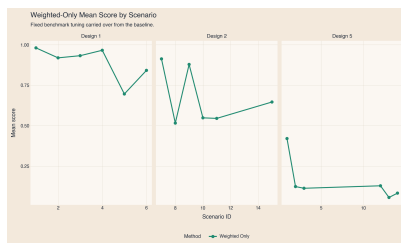
Figure 2: Complete baseline tuning frontiers. Moderate aggregation thresholds and a repetition budget of 1000 provide the most useful overall compromise.

4.2 Weighted aggregation

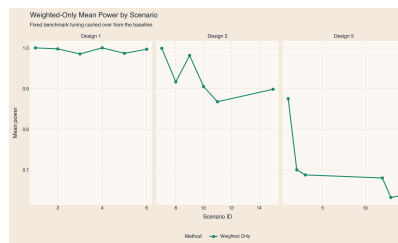
The weighted-only study changes only the aggregation layer. Across 18 conditions, it attains mean score 0.572, power 0.875, error 0.302, and runtime 451.62 seconds. The method performs well on Design 1 but becomes overly aggressive on Design 5, where mean error reaches 0.548. Table 3 and Figures 3–4 show all weighted results.

Table 3: Design-wise weighted-only results.

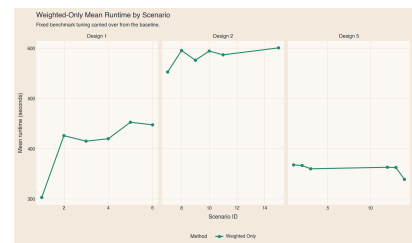
Design	Score	SD score	Power	Error	Runtime (s)	Model size
Design 1	0.889	0.107	0.994	0.105	410.62	4.90
Design 2	0.674	0.177	0.928	0.254	584.52	9.51
Design 5	0.154	0.133	0.702	0.548	359.71	7.71



(a) Mean score by scenario



(b) Mean power by scenario



(c) Mean runtime by scenario

Figure 3: Weighted-only scenario results. The strong power profile is accompanied by weaker error control on the most difficult design.

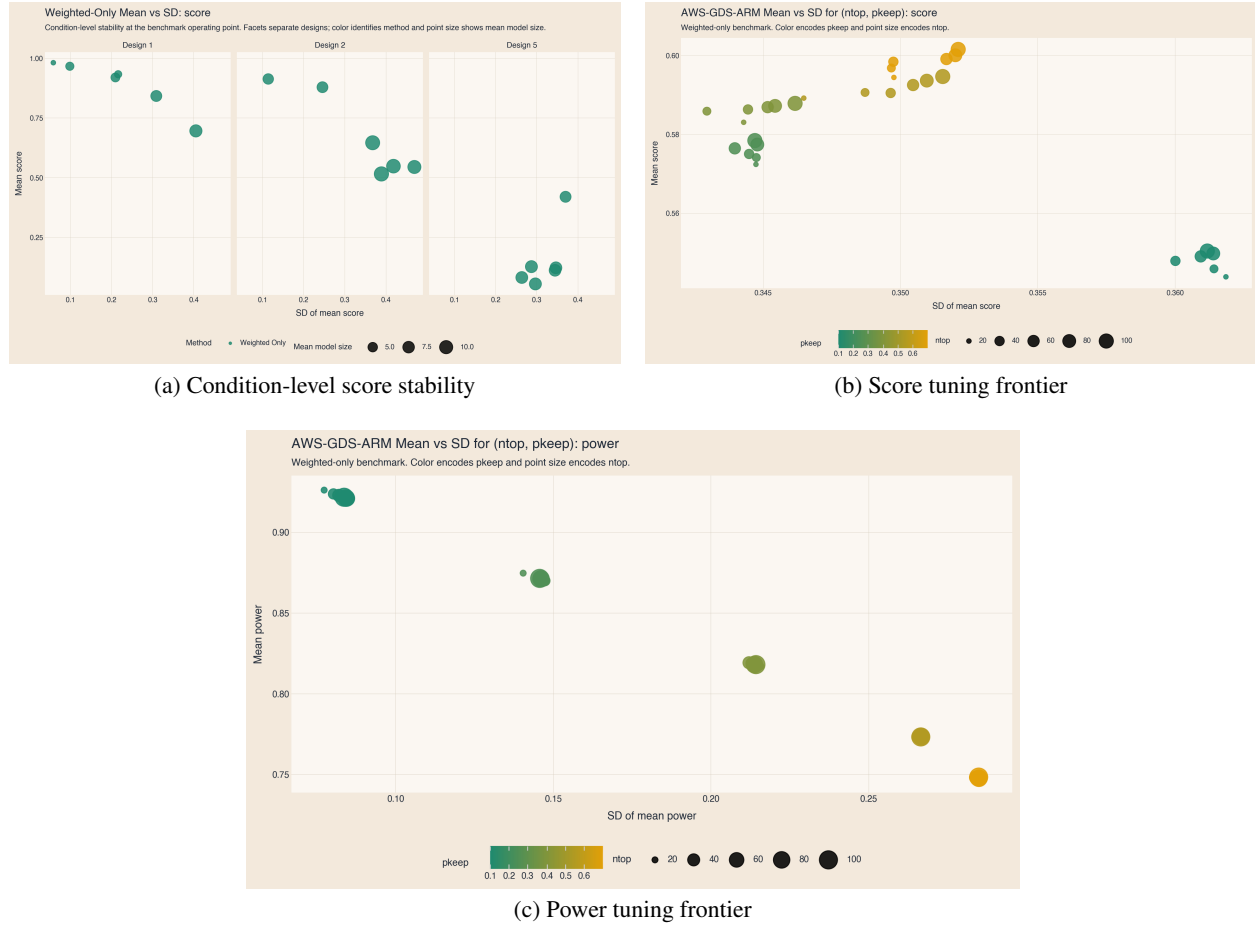


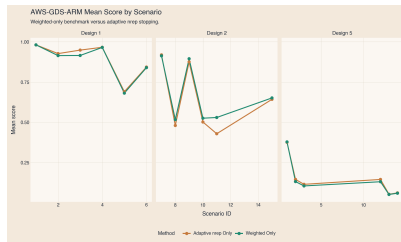
Figure 4: Weighted aggregation stability and tuning results. Settings that maximize power do not necessarily maximize the combined score.

4.3 Adaptive repetition stopping

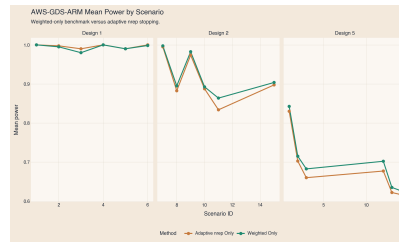
Adaptive stopping uses 368.81 repetitions on average and stops early in 92.5% of runs. In the matched ablation bundle, runtime falls from 388.06 to 137.96 seconds, a reported 3.27-fold speedup, while score changes from 0.566 to 0.561. Design 2 shows the largest score cost; Designs 1 and 5 show small score gains. Table 4 and Figures 5–6 contain all stopping results.

Table 4: Adaptive-stopping results. The upper rows give the overall ablation; the lower rows give design-wise changes relative to weighted aggregation.

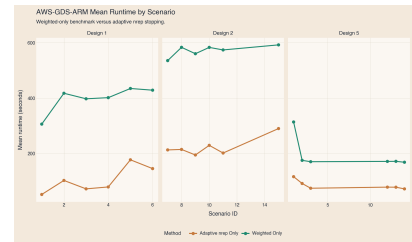
Comparison	Repetitions	Stop rate	Score	Power	Error	Runtime (s)
Weighted overall	1000.00	0.000	0.566	0.872	0.307	388.06
Adaptive overall	368.81	0.925	0.561	0.864	0.303	137.96
Design	Repetitions	Stop rate	Weighted score	Adaptive score	Change	Speedup
Design 1	253.38	0.972	0.883	0.892	0.010	3.80
Design 2	390.67	0.958	0.672	0.641	-0.031	2.55
Design 5	462.38	0.846	0.143	0.149	0.006	2.29



(a) Mean score by scenario

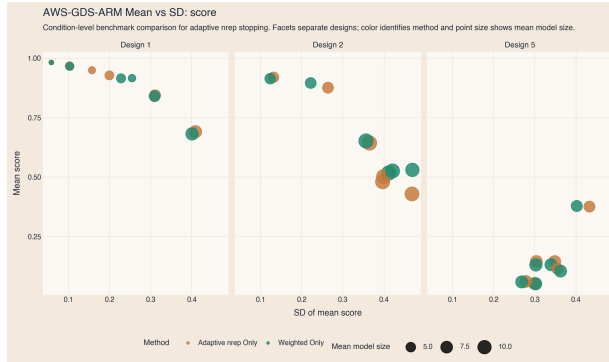


(b) Mean power by scenario

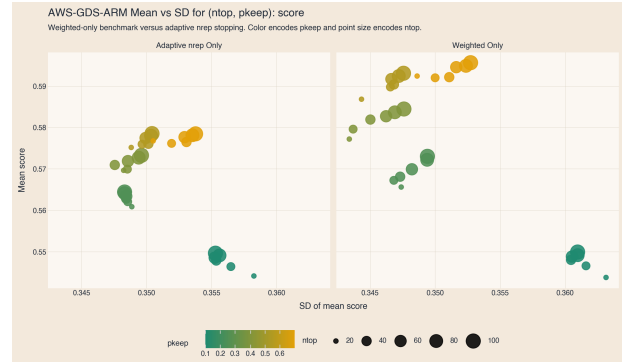


(c) Mean runtime by scenario

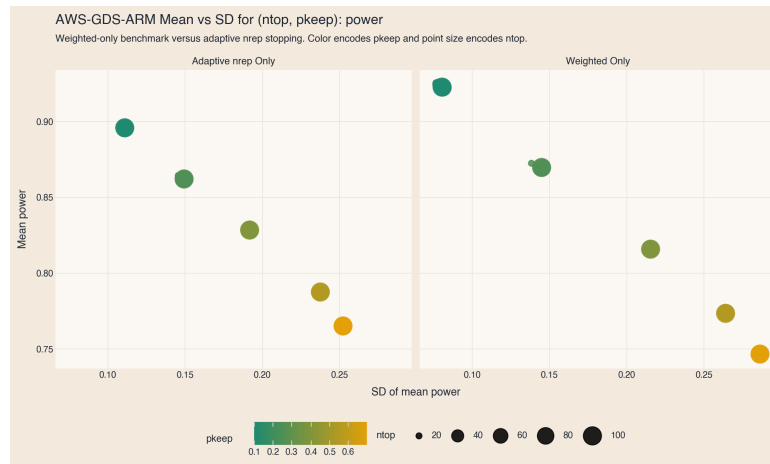
Figure 5: Adaptive-stopping scenario comparison. Selection performance stays close to the weighted benchmark while runtime falls across most scenarios.



(a) Condition-level score stability



(b) Score tuning frontier



(c) Power tuning frontier

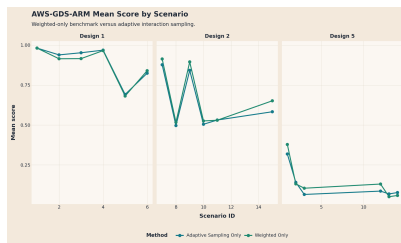
Figure 6: Adaptive-stopping stability and tuning results. Early stopping primarily changes cost rather than the underlying weighted selection frontier.

4.4 Adaptive interaction sampling

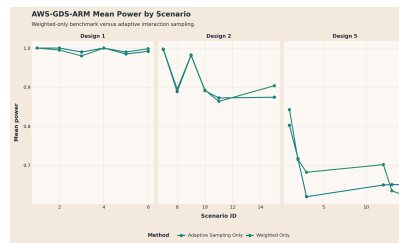
Adaptive sampling keeps weighted aggregation and $nrep = 1000$ fixed. Overall score is 0.552 versus 0.566 for weighted uniform sampling; runtime is 221.46 versus 216.79 seconds. The adaptive proposal helps Design 1 slightly but loses score on Designs 2 and 5. Table 5 and Figures 7–8 report all sampling results.

Table 5: Adaptive-sampling results. Changes are adaptive minus weighted.

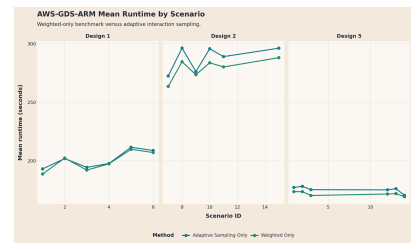
Comparison	Score	Power	Error	Runtime (s)	Runtime ratio
Weighted overall	0.566	0.872	0.307	216.79	1.00
Adaptive overall	0.552	0.865	0.312	221.46	1.02
Design	Weighted score	Adaptive score	Score change	Error change	Runtime ratio
Design 1	0.883	0.892	0.010	-0.009	1.01
Design 2	0.672	0.639	-0.033	0.028	1.03
Design 5	0.143	0.126	-0.016	-0.002	1.02



(a) Mean score by scenario



(b) Mean power by scenario



(c) Mean runtime by scenario

Figure 7: Adaptive-sampling scenario comparison. Gains are localized and do not shift the overall score frontier upward.

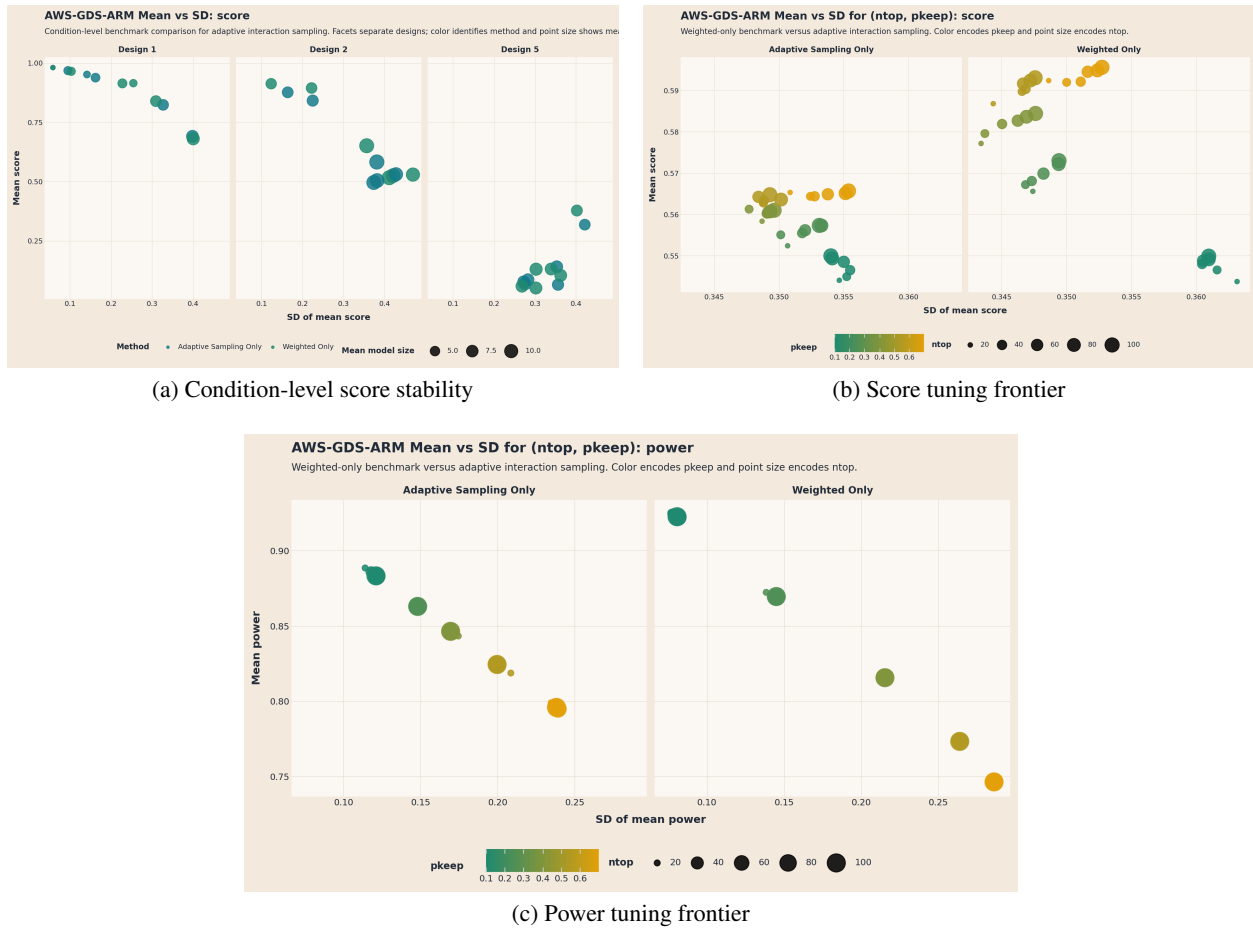


Figure 8: Adaptive-sampling stability and tuning results. The proposal mechanism remains sensitive to design and threshold choice.

4.5 Unified four-method comparison

The unified study is the decisive comparison because all four methods share one configuration. The original baseline attains the best mean score, 0.601, and the lowest mean error, 0.235. Weighted aggregation has the highest power, 0.872, but increases error to 0.307. Adaptive stopping stays close to the weighted method on score while reducing mean runtime by 63%. Adaptive sampling is slightly slower than the original and has the lowest mean score.

Table 6: Overall four-method comparison under common settings.

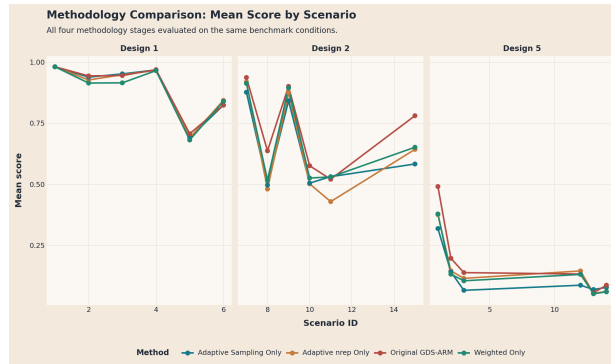
Method	Score	Power	Error	Runtime (s)	Model size	Repetitions
Original baseline	0.601	0.836	0.235	216.22	6.04	1000.00
Weighted aggregation	0.566	0.872	0.307	216.21	7.43	1000.00
Adaptive stopping	0.561	0.864	0.303	79.93	7.27	368.81
Adaptive sampling	0.552	0.865	0.312	220.77	7.28	1000.00

The baseline wins score in 12 of 18 conditions, including five of six on Design 2 and four of six on Design 5. Weighted aggregation never wins score but wins power in six conditions. Adaptive sampling has three score wins and five power wins. Adaptive stopping has three score wins and is the runtime winner in all 18 conditions.

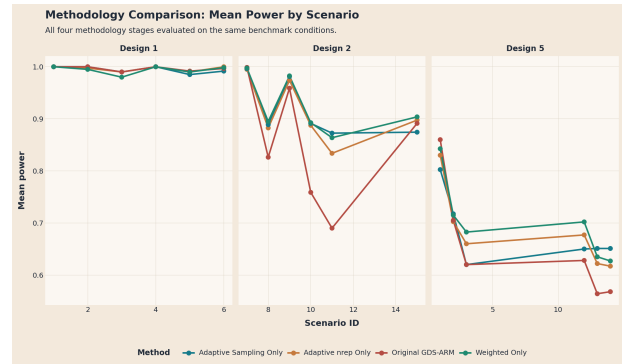
Table 7: Condition-level wins. Entries list original / weighted / adaptive stopping / adaptive sampling.

Design	Score wins	Power wins	Runtime winner
Design 1	3 / 0 / 2 / 1	3 / 0 / 2 / 1	Adaptive stopping (6/6)
Design 2	5 / 0 / 0 / 1	1 / 4 / 0 / 1	Adaptive stopping (6/6)
Design 5	4 / 0 / 1 / 1	1 / 2 / 0 / 3	Adaptive stopping (6/6)

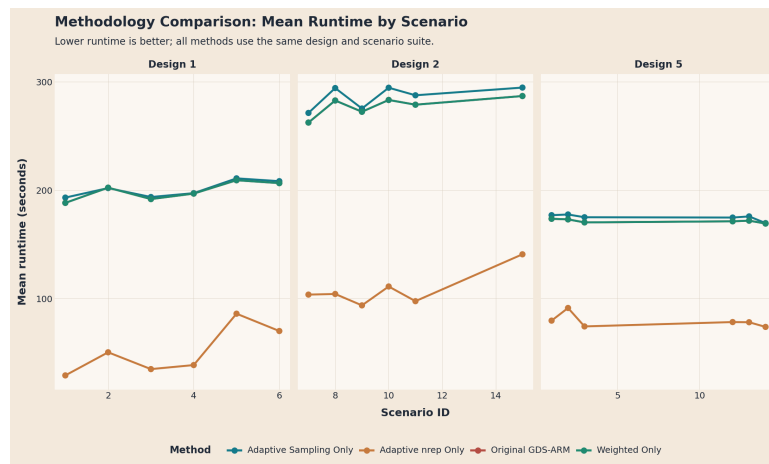
Figures 9–12 present every unified-comparison plot. Together they show the governing tradeoff: the adaptive and weighted variants often gain power by selecting larger models and accepting more false positives, while adaptive stopping alone creates a clear shift in computational cost.



(a) Mean score by scenario



(b) Mean power by scenario



(c) Mean runtime by scenario

Figure 9: Unified scenario comparison. The baseline leads score across much of Designs 2 and 5, while adaptive stopping leads runtime throughout.

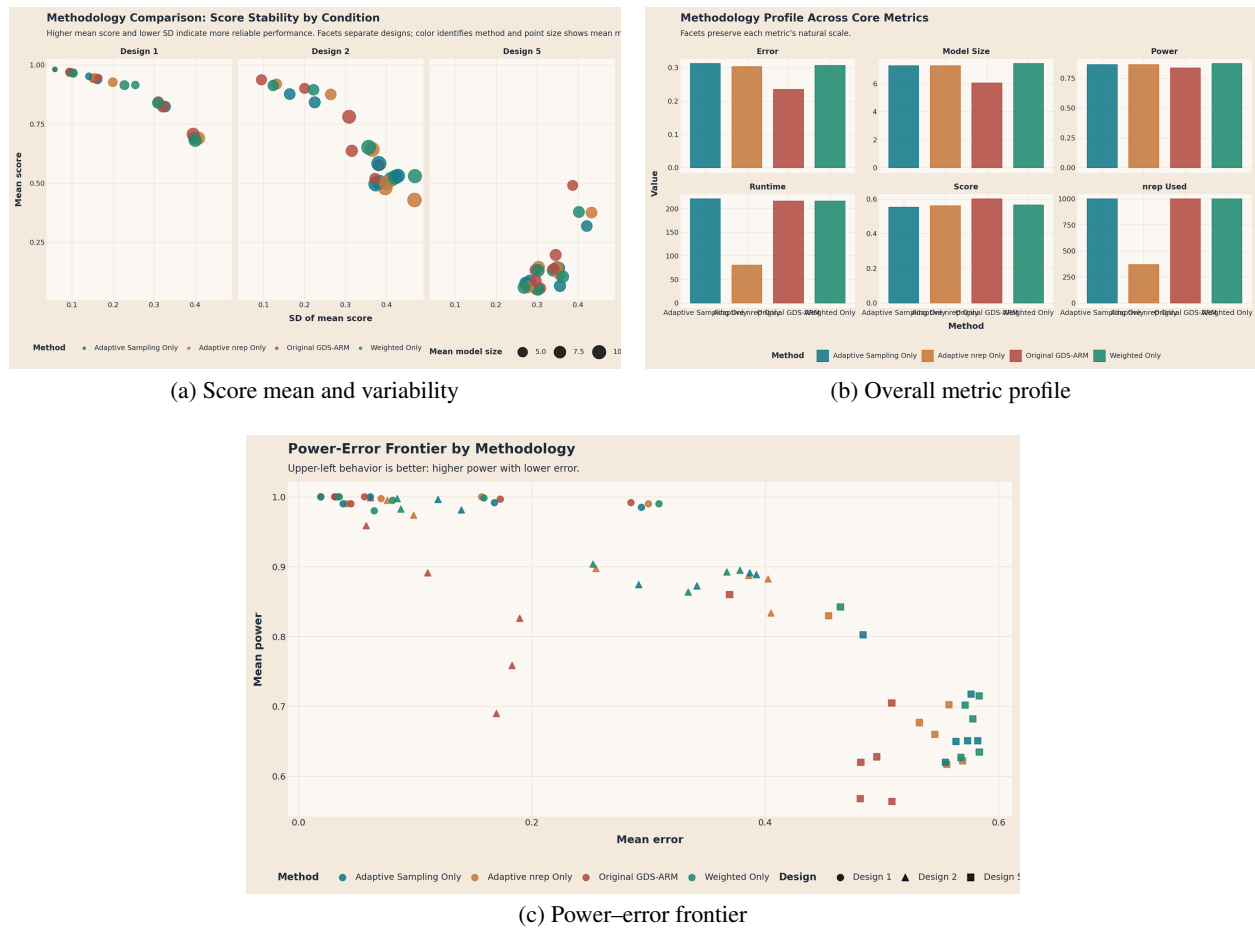


Figure 10: Unified selection-quality views. Higher power from the weighted and adaptive variants is generally accompanied by higher false-selection error.

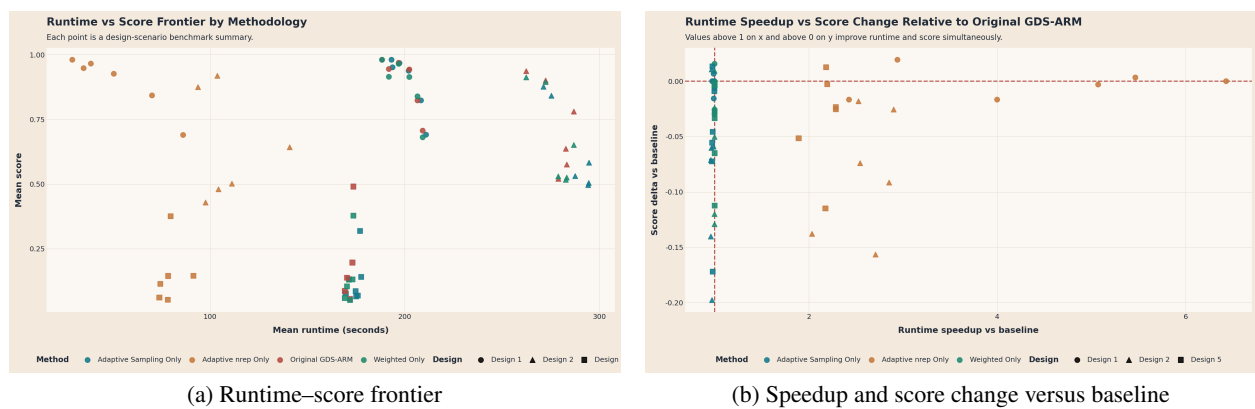
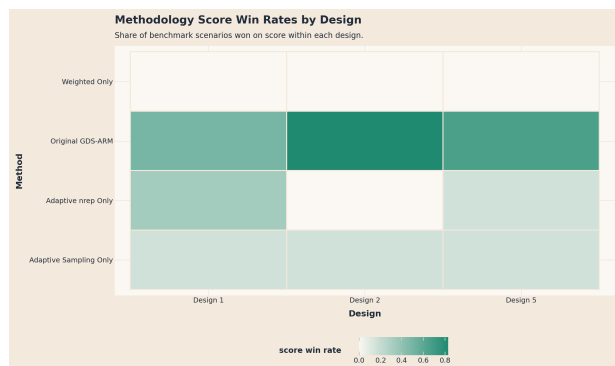
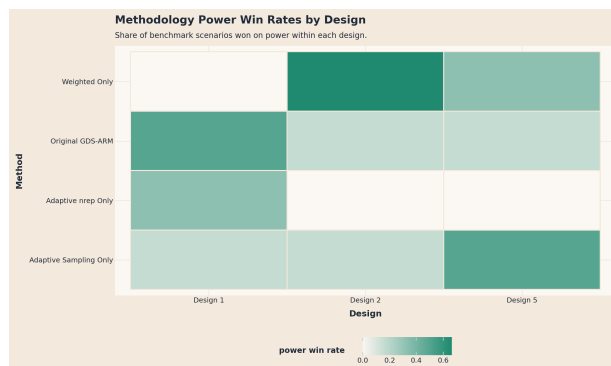


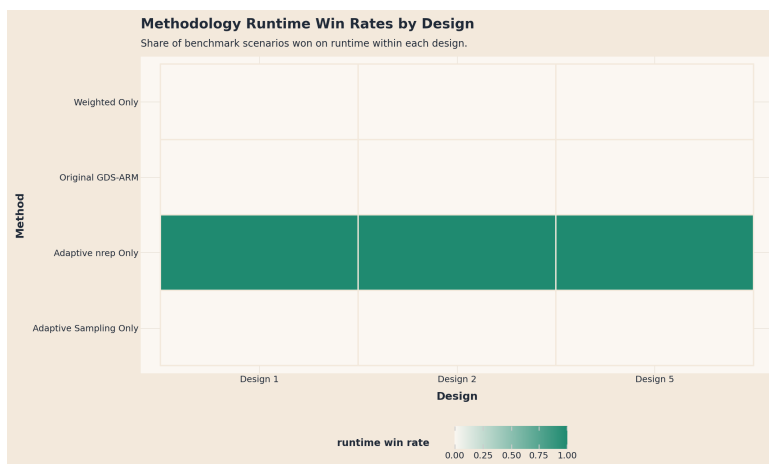
Figure 11: Unified computational tradeoffs. Adaptive stopping provides the only systematic runtime improvement, although most conditions do not also improve score.



(a) Score win rates



(b) Power win rates



(c) Runtime win rates

Figure 12: Win-rate summaries by design. The original baseline dominates score on Designs 2 and 5; adaptive stopping wins every runtime comparison.

5 Discussion

The results separate the three adaptive ideas clearly. BIC weighting changes the selection frontier: it raises sensitivity, enlarges the selected model, and increases false selections. Adaptive stopping changes computational effort while leaving the weighted frontier largely intact. Adaptive interaction sampling changes where the search is concentrated, but its current tuning does not improve the global power–error balance.

If the combined score is the primary criterion, the original mixed-level GDS-ARM baseline remains preferred. Its advantage is strongest on Designs 2 and 5, where error control matters more than marginal power gains. If runtime is the binding constraint, adaptive stopping is the strongest modification: it reduces the repetition budget by about 63% under the common comparison and wins every condition on runtime.

Practical Interpretation

Use the original baseline for the best overall score and smallest false-selection rate. Use adaptive stopping when a modest change in the weighted score is acceptable in exchange for a substantial reduction in computation. The weighted and adaptive-sampling rules require additional threshold or design-specific tuning before they can replace the baseline.

Two limitations qualify the numerical conclusions. First, the design matrices reproduce the intended dimensional templates and coding, not the exact supplement rows. The implementation is therefore algorithmically faithful but only a qualitative numerical reproduction. Second, the weighted threshold and adaptive-sampling controls were not tuned separately by design class. Future work should use the exact design rows, tune error control explicitly, and test a hybrid procedure that combines baseline aggregation with sequential stopping.

6 Conclusion

The original GDS-ARM baseline remains the strongest overall factor-screening method in the reported mixed-level benchmark. Weighted aggregation improves power but gives up too much false-selection control, and adaptive interaction sampling produces only localized benefits. Adaptive repetition stopping is the most useful extension because it sharply reduces computation while preserving most of the weighted method’s selection behavior. The central contribution is therefore a modular conclusion: adaptivity is already valuable for stopping, while adaptive aggregation and sampling still require refinement.

References

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- [2] Phoa FK, Pan YH, Xu H (2009) Analysis of supersaturated designs via the Dantzig selector. *Journal of Statistical Planning and Inference* 139(7):2362–2372
- [3] Singh R, Stufken J (2024) Factor selection in screening experiments by aggregation over random models. *Computational Statistics & Data Analysis* 194:107940
- [4] Zhang F, Singh R, Stufken J (2026) An extension of the GDS-ARM algorithm for factor screening in mixed-level supersaturated designs. *Journal of Statistical Theory and Practice* 20:41. <https://doi.org/10.1007/s42519-026-00542-x>